1) Show that $\frac{8+\sqrt{12}}{5+\sqrt{3}}$ can be written in the form $\frac{a+\sqrt{3}}{b}$, where a and b are integers.
[Edexcel - Nov 2021]

2) Show that $\frac{\sqrt{180}-2\sqrt{5}}{5\sqrt{5}-5}$ can be written in the form $a+\frac{\sqrt{5}}{b}$, where $a\ and\ b$ are integers.
[Edexcel – May 2020]

3) a) Rationalise the denominator of $\frac{22}{\sqrt{11}}$. Give your answer in its simplest form.

b) Show that $\frac{\sqrt{3}}{2\sqrt{3}-1}$ can be written in the form $\frac{a+\sqrt{3}}{b}$, where a and b are integers.

4) a) Express $\sqrt{3}+\sqrt{12}$ in the form $a\sqrt{3}$, where a is an integer.

b) Express $\left(\frac{1}{\sqrt{3}}\right)^7$ in the form $\frac{\sqrt{b}}{c}$, where b and c are integers. **[Edexcel – June 2019]**

5) Show that $\frac{\left(\sqrt{18}+\sqrt{2}\right)^2}{\sqrt{8}-2}$ can be written in the form $a(b+\sqrt{2})$ where a and b are integers.

6) Show that

$$\frac{1}{1+\frac{1}{\sqrt{2}}}$$

Can be written as $2-\sqrt{2}$.

7) Express $\sqrt{48}+\sqrt{108}$ in the form $k\sqrt{6}$ where k is a surd.

8) Show that $\sqrt{27}+\sqrt{147}$ can be expressed in the form $a\sqrt{b}$, where a and b are integers.

9) Simplify $(7+2\sqrt{50})(5-2\sqrt{2})$. Give your answer in the form $a+b\sqrt{18}$ where a and b are integers. Show your working clearly.

10) Show that $\frac{\sqrt{3}+\sqrt{27}}{\sqrt{2}}$ can be expressed in the form $\sqrt{k}$ where k is an integer. State the value of k.

11) a) Expand $\left(5+3\sqrt{2}\right)^2$. Give your answer in the form $\left(a+b\sqrt{2}\right)$, where a and b are integers. Show your working clearly.

b) $\left(5+3\sqrt{2}\right)^2=p+\frac{q}{\sqrt{8}}$, where p and q are integers. Find the value of q.

12) Given that $\left(5-\sqrt{x}\right)^2=y-20\sqrt{2}$ where x and y are positive integers, find the value of x and the value of y.

13) $(3+\sqrt{a})(4+\sqrt{a}) = 17 + k\sqrt{a}$ where a and k are positive integers, find the value of a and k.

14) $(\sqrt{a}+\sqrt{8a})^2 = 54 + b\sqrt{2}$. Where a and b are positive integers. Find the value of a and the value of b. Show your working clearly.

15) $(a+\sqrt{b})^2 = 49 + 12\sqrt{b}$, where a and b and integers, and b is a prime number. Find the values of a and b.

16) Simplify fully

$$\frac{(6-\sqrt{5})(6+\sqrt{5})}{\sqrt{13}}$$

You must show all your working.

17) Rationalise the denominator and simplify fully

$$\frac{33}{4+\sqrt{5}}$$

Show clear working out.

18) Express

$$\frac{39}{4-\sqrt{3}}$$

In the form $a + b\sqrt{3}$, where a and b are integers. Show clear working out.

19) Simplify

$$\frac{7-\sqrt{5}}{2+\sqrt{5}}$$

Giving your answer in the form $a + b\sqrt{5}$, where a and b are integers. Show clear working out.

20) Show that

$$\frac{3}{\sqrt{27}-\sqrt{18}}$$

Can be written in the form $\sqrt{m}+\sqrt{n}$, where m and n are integers.

21) Show that

$$\frac{16}{\sqrt{2}} - \sqrt{8} = 6\sqrt{2}$$